

TOPIC 2.A

Aims, tools, and problems of nature conservation

1. What is the aim of nature conservation? What legal, economic, and educational (awareness-raising) options exist for the protection of species, areas (biological communities), and unique natural values?
2. What active intervention can have a beneficial nature conservation effect (2 examples)?
3. Describe a national park in Hungary! What is its main value (three examples)? What problems and difficulties must nature conservation overcome in this area? What solutions can be found for the listed problems (2 examples)?
4. What genetic dangers threaten populations that are too small in number? How can this be prevented?
5. Analyse an arbitrary domestic case in which the interests of nature conservation conflicted with other (e.g., economic) interests! How did the debating parties argue in this case? Whose interest prevailed? Who would you agree with? What would you suggest as a solution?

Expected knowledge	Attainable score	Achieved score
1. • Formulating the aim of nature conservation • Protection of species: legally: e.g., prohibition, conventions; economically: e.g., compensation for damage; through emotion and awareness-raising: e.g., teaching, campaigns. • Protection of biological communities / biological diversity: protected areas, landscape protection areas, national parks. • Unique values: local protection: e.g., old trees.	6 points	
2. • Examples of active nature conservation: making bird nest boxes / frog rescue / regular mowing / flooding of wetlands • captive breeding in zoos and reintroduction / suppression of invasive species, etc. at least two examples.	2 points	
3. • Example of a National Park (naming), • its characteristic types of biological communities and other values, • e.g., Aggtelek: karst region, dripstone cave / cave biota • on its surface limestone rock grasslands (three characteristics, these can also be names of protected species or biological communities) • Description of two problems, e.g., contamination of infiltrating waters in the karst region, invasion of non-native species • and their solutions, e.g., waterworks protection zones / sewerage and biological wastewater treatment plants, and • the suppression of non-native tree species, planting of native ones.	8 points	

4. • Genetic dangers: inbreeding depression and the irreversible effect of random changes. • Prevention: contiguous protected area of appropriate size / ecological corridors. • Mentioning another method, e.g., artificial breeding, is also acceptable.	3 points	
5. Case analysis, forming an opinion: • Description of the case. E.g., the expansion of the cement factory is planned in the village. • formulating the nature of the conservation damage, e.g., it involves the destruction of protected rock grassland. • summarising the arguments and counterarguments, e.g., it creates jobs, there will be new filtering equipment, but: it destroys protected grasslands, threatens the quality of local vineyards, tourism decreases, etc. • based on the weighing of advantages and disadvantages, forming an opinion, formulating a suggestion.	6 points	
5. Structure of the answer, communication skills	5 points	
Total:	30 points	

TOPIC 2.B

Comparison of a moss and a fern plant

To complete the tasks, use the Plant Identification book!

1. From the plants on the tray, select the moss and the fern! Name the exact species names!
2. Identify the fern plant! Based on which visible characteristic did you classify it into the phylum of ferns (Pteridophytes)?
3. Characterise its ecological requirements (heat, nitrogen, and water requirements, disturbance tolerance)! In which biological community (association) can we find it?
4. Characterise the structure of mosses and ferns through the two plants! Name the main differences! In what way is their reproduction similar?
5. How do mosses adapt to changing water conditions? What is their significance in the life of forest biological communities, in the protection of the soil?



Expected knowledge	Attainable score	Achieved score
1. Naming the two species.	2 points	
2. • Recognition of the fern - maidenhair spleenwort (<i>Asplenium trichomanes</i>) - (based on the description / photograph in the Plant Identification book). • Based on the location of the sporangia or: it has true leaves, but it does not have flowers.	2 points	
3. • Characterisation of the habitat of the maidenhair spleenwort based on the TWR table of Plant Identification (p. 251). • (water requirement: 6, heat requirement: 0, nitrogen content of the environment: not char., disturbance tolerance: 3) - description of four characteristics (2 points), • it is a denizen of shady rock grasslands.	3 points	
4. Phylum of mosses (Bryophytes), ferns (Pteridophytes): • Mosses: thalloid plant, • generally lacks tissues, organs. • Ferns: appearance of tissues: - epidermal tissue: protects against desiccation, - vascular tissue: transport of	10 points	

water and organic matter, - supporting tissue: supporting the plant. • Organs: true root, stem, leaf. • Members of both groups reproduce by spores, fertilisation is bound to water.		
5. • Mosses are poikilohydric plants (plants with varying water states): they dry out, and then they are able to absorb an amount of water corresponding to many times their body mass. • They occur in almost all habitats due to their wide tolerance range. • Their role: binding of precipitation water / reduction of soil erosion.	3 points	
Total:	20 points	